SUSTAINABILITY

Operationalizing the OCP Spec for Data Center Equipment: **LCA at Scale and the Open Enablers**

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OCP PLATINUM Member

Problem Statement

To drive reductions, the industry needs **transparent**, **granular**, **high-fidelity** and **actionable** Product Carbon Footprints (PCFs).

CHALLENGE 01

Methodologies

Inconsistent Life Cycle Assessment (LCA) and carbon accounting methodologies make consistency and comparisons difficult.

CHALLENGE 02

Actionability

Spend-based emissions often must be used at scale, but don't effectively quantify reductions.

CHALLENGE 03

Transparency

A lack of supply chain transparency (e.g. manufacturing data) persists across the industry due to IP concerns, reducing actionability.

The scale and complexity of AI systems has amplified these challenges.

Our Strategy

Product Category Rules

Funded PCRs or OCP Specifications for:

- Consumer devices (Laptops & PCs) via OCP
- Data Center Equipment via OCP
- Wafer-level via SEMI

LCA at Scale

A framework for applying Life Cycle Assessment (LCA) at scale across complex Information and Communications Technology (ICT) hardware portfolios drawing on the collective learnings of the hyperscalers.

Open Enablers Program

Data and software enablers to accelerate decarbonisation progress across the ICT hardware industry.

The foundation for LCA at Scale: the **OCP Spec**

Component Emissions Foundation

1. Taxonomy of 'What's in a Rack'

Enables granular categorization and hotspot analysis across systems.

2. Component Modelling Rules

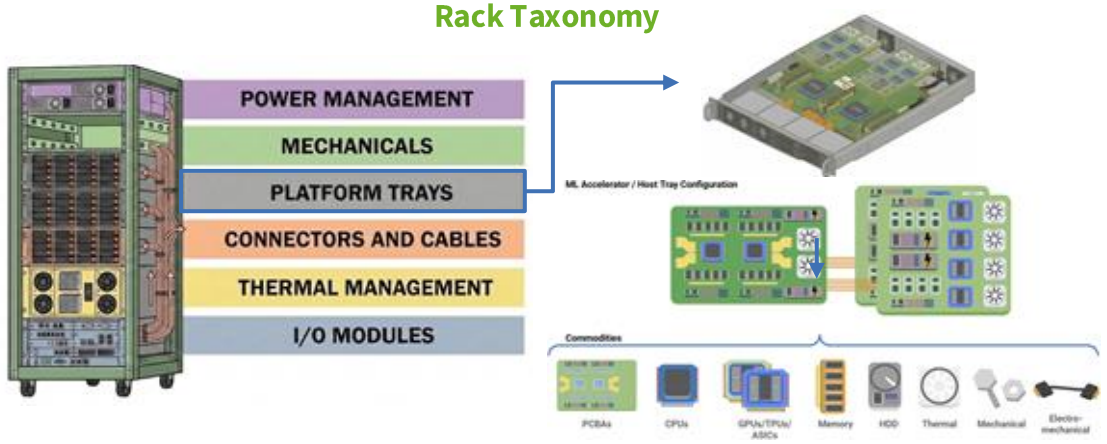
Establishes consistent carbon accounting rules for key, high-impact components.

3. Data Quality Scoring (DQS)

Built on the PACT[1] framework to enable scalable Data Quality Scoring on a 1-10 scale for complex assemblies and product portfolios.

[1] https://www.carbon-transparency.org/

Rack Taxonomy



Data Quality Score

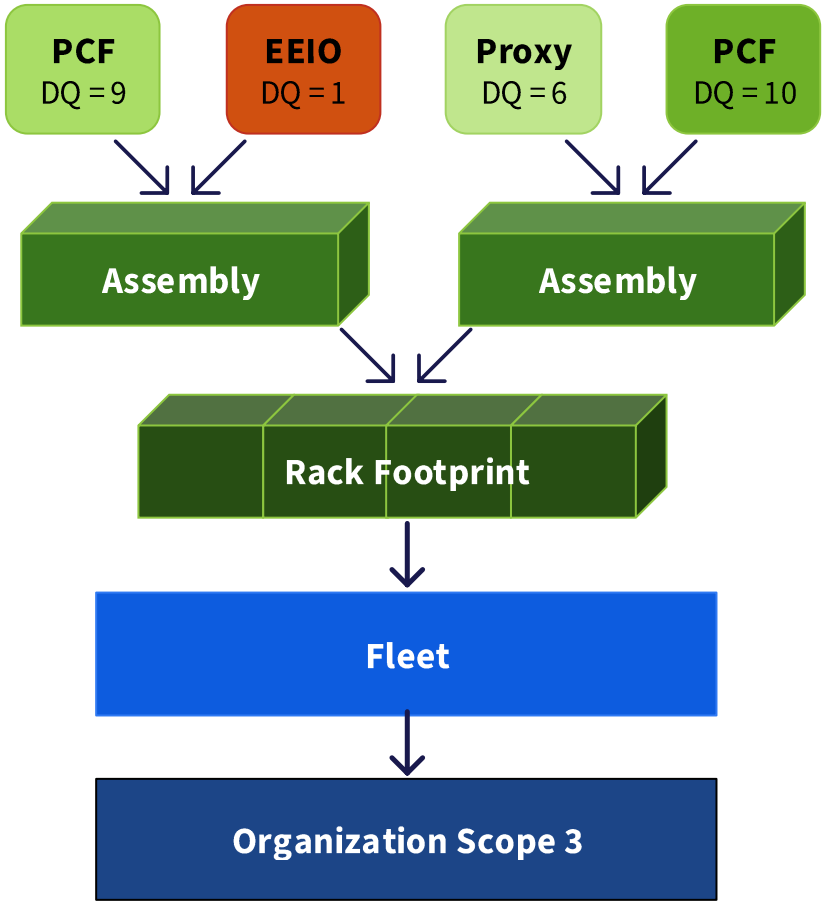
Mechanical scoring

- Validate PCF prerequisites
- Qualitative PCF scoring
- Propagate scores

Score	Input Type
8–10	PCFs: Primary-data PCF
5-7	Proxy PCF
4	Parametric Model: Generic Inputs
3	Category Average
2	Supplier Emission Factor (Revenue-Based)
1	Price-Based EEIO (Environmentally Extended Input-Output)
0	No data

LCA at Scale Framework Overview

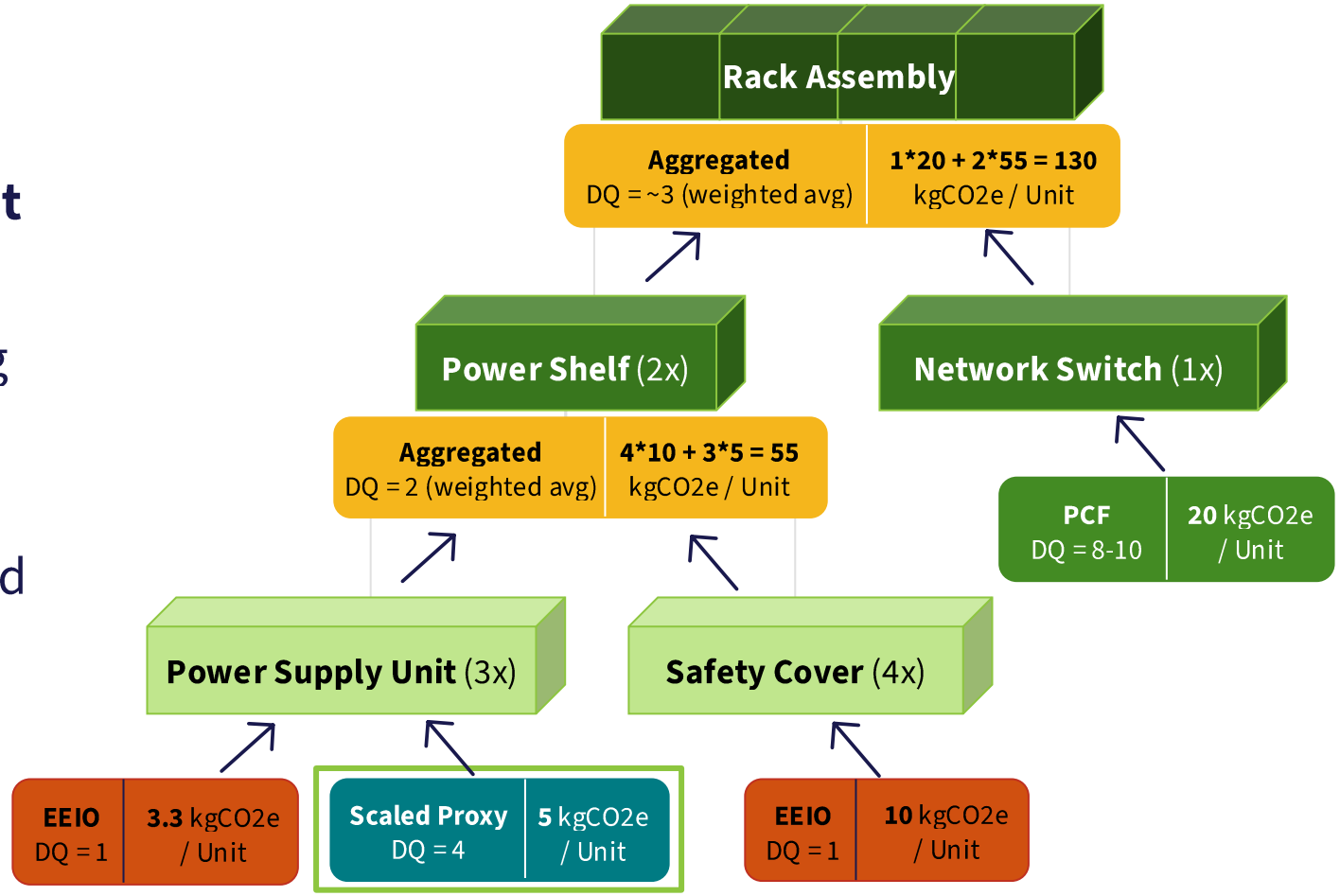
- **Component emissions:** data quality driven emissions selection
- **Product Carbon Footprints (PCFs):** using the Bill of Materials (BOM), aggregate component emissions and data quality to the product/assembly level
- **Corporate Scope 3 Reporting:** scale the PCFs to the entire product portfolio, using procured quantities.



Scaling **Component Emissions** to **PCFs**

Scale PCFs for assemblies & products

1. For each component: **select the highest quality emissions estimate**
2. **Aggregate emissions** bottoms up using the Bill of Materials for each sub-assembly and product configuration
3. **Calculate data quality** using a weighted average of subordinate data quality.



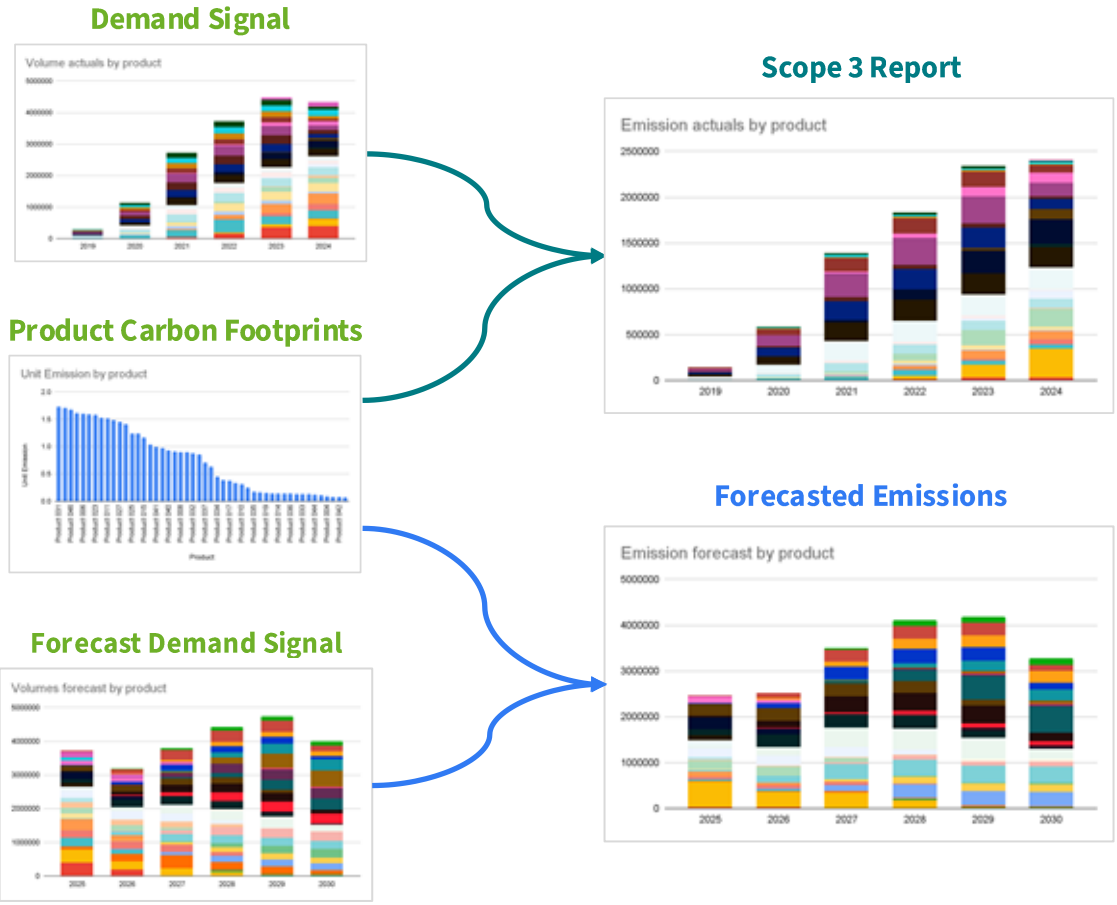
Scaling to **Scope 3 Reporting** and **Forecasting**

Scale PCFs to Corporate Scope 3 Reporting

- Join component, assembly and product emissions to **actuals** demand signal
  - PO receipts for upstream Scope 3
  - Sales orders & shipments for downstream Scope 3

Scale PCFs to Forecast emissions

- Join to **forecast** demand signal
  - Deployment forecasts
  - Sales forecasts



* Data is for illustrative purposes only

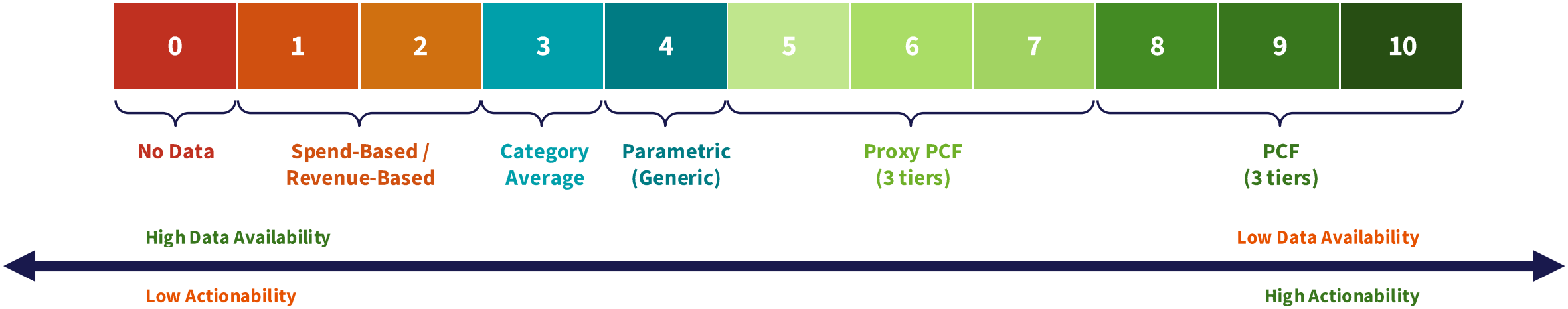
The completeness and quality paradox

The Paradox

Organizations must produce credible, complete Scope 3 reports today while transitioning from low-fidelity, spend-based estimates toward comprehensive LCA models.

The Solution

The **LCA at Scale** framework enables combining emissions of varying data qualities, to prioritize high-fidelity LCAs where it matters and leverage lower quality estimates for completeness.

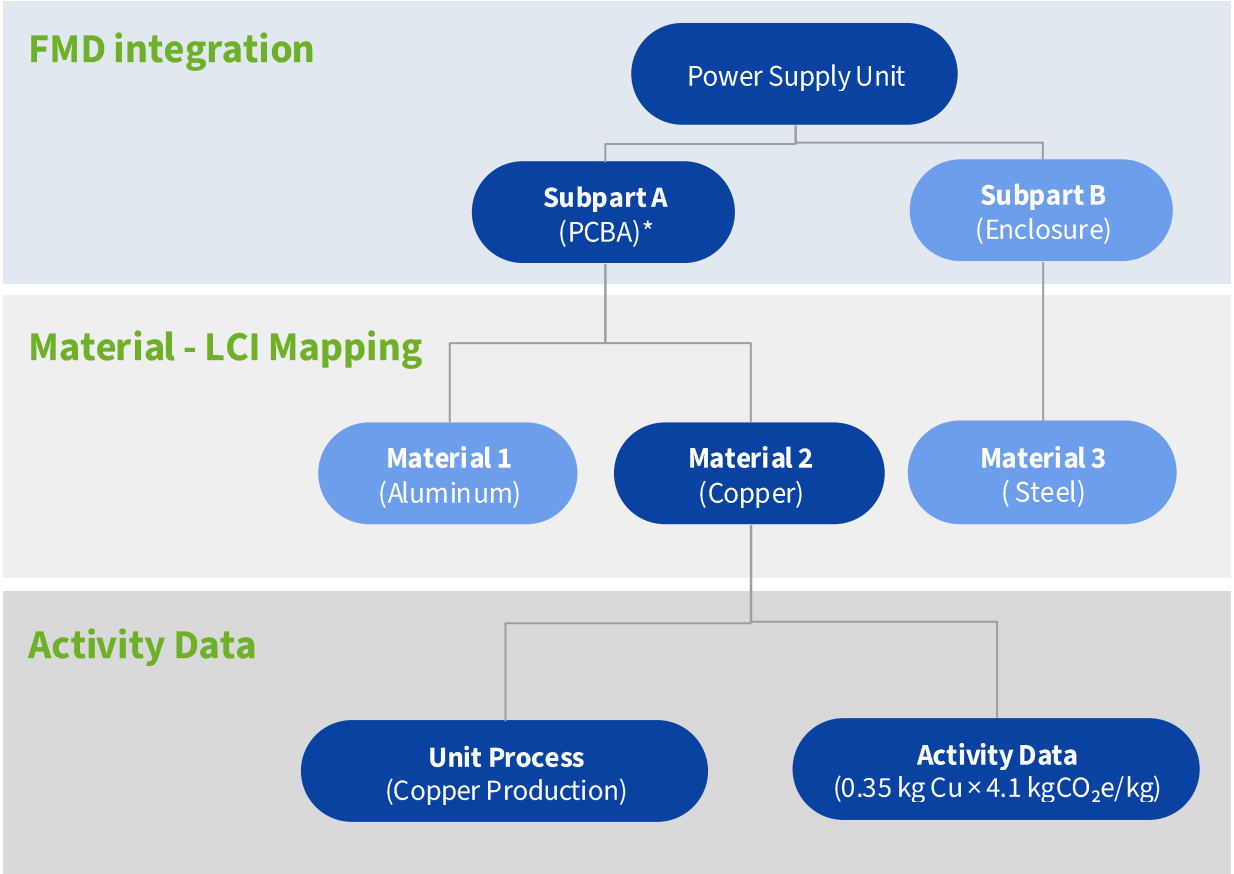


Case Study: **Full Material Disclosures**

Full Material Disclosures (**FMDs**) can be leveraged for compliance and granular product inventory for high-fidelity LCAs.

HOW IT WORKS

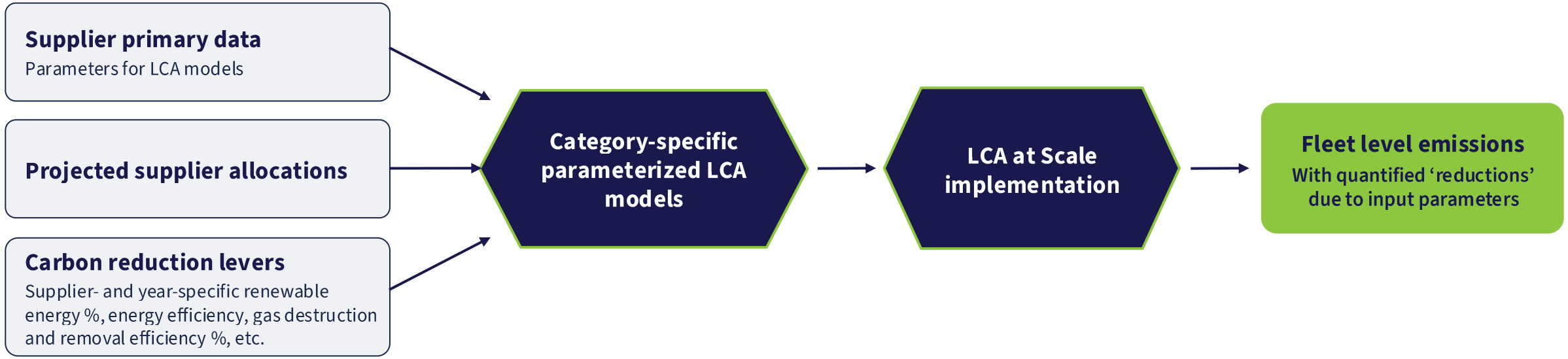
1. Exchange FMDs throughout the supply chain
2. AI-assisted FMD parsing and integration at the component level
3. LCA calculated with material to lifecycle inventory (LCI) mapping and more detailed activity data.



* PCB Assembly and Test emissions are not included in the diagram

Case Study: **Parameterized LCAs and emissions reduction modelling**

Parameterized LCAs can be used with supplier primary data for actionable LCAs and supplier-specific emissions reduction modelling.

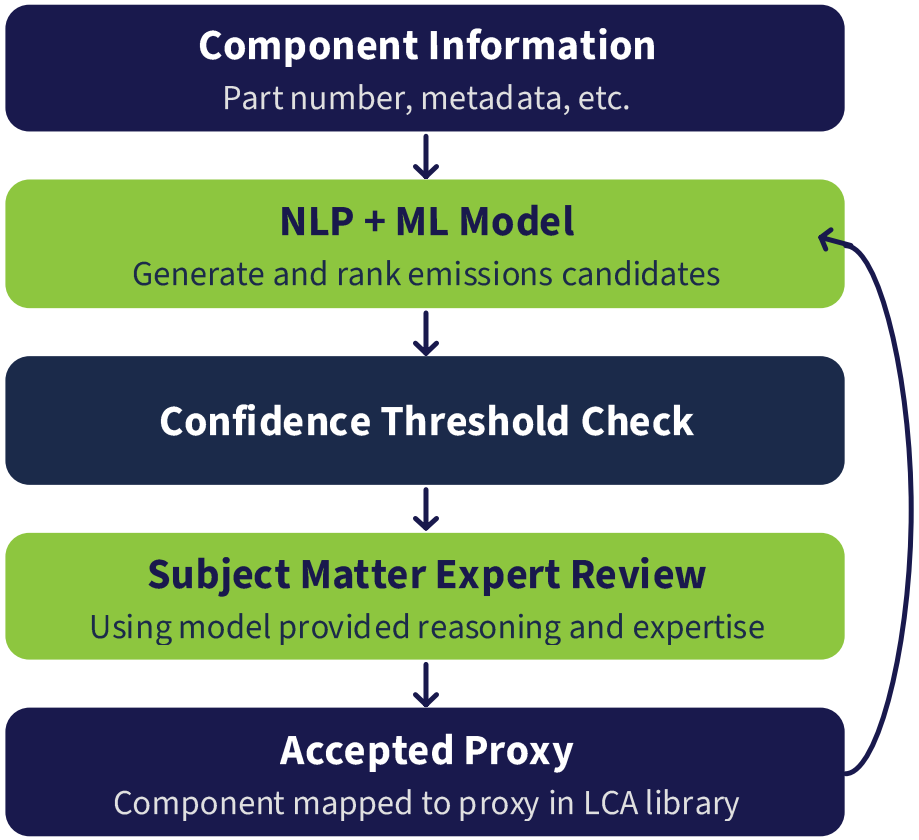


Case Study: **AI-Assisted Proxy Matching**

Manual emissions assignment is the bottleneck to LCA coverage completeness.

AI-ASSISTED WORKFLOW

- Build domain-knowledge library based on hardware physical properties (DDR or PCIe generation, fabrication processes, etc.)
- Leverage AI to propose PCF proxies from similar components based on metadata (category, description, properties, etc.)
- All proxy proposals are both automatically and manually validated
- Review patterns become new rules embedded into the model.



Open Enablers: Enhancing Completeness for the Industry

WHAT

An OCP initiative to create a living component emissions dataset of industry average Data Center Equipment PCF values which doesn't compromise the confidentiality/IP of PCF contributors.

Category	Declared Unit	Industry Average PCF
Hard Disk	18TB	50kg
CPU	32-64 cores, Chiplet Architecture	100kg

* Data is for illustrative purposes only

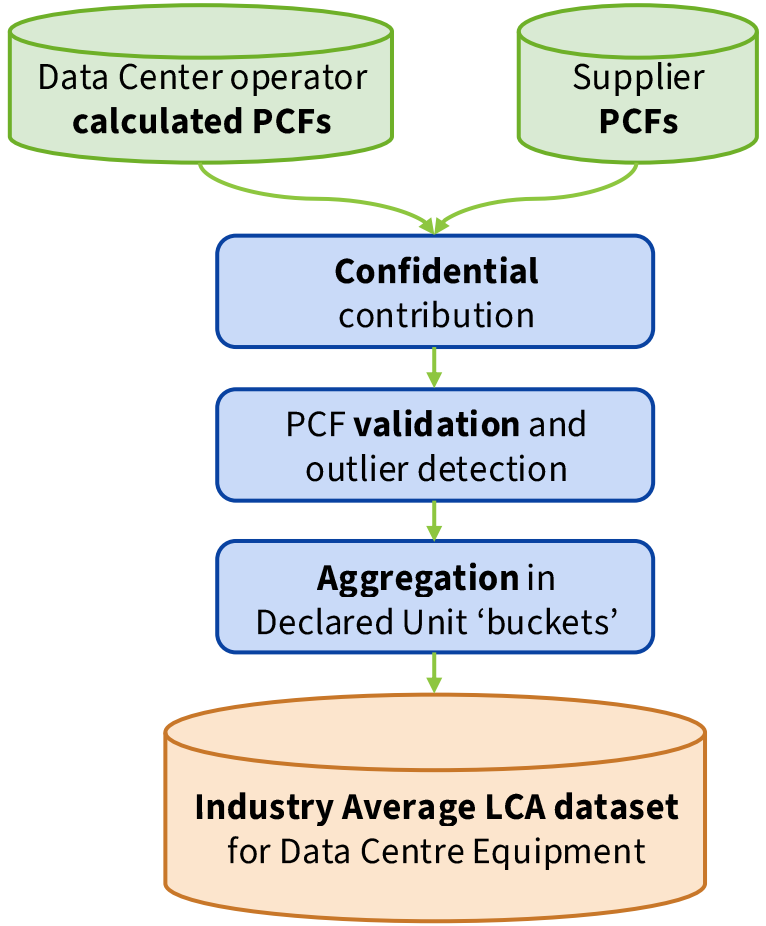
USE CASES

- **Screening level LCA** for complex hardware assemblies
- **Gap filling** where supplier data is not available, reducing the margin of error relative to spend-based emissions
- **Quality assurance** against high quality, up to date, Industry Average data.

Open Enablers: Enhancing Completeness for the Industry

HOW IT WORKS

- **Anonymized PCF contributions** to a 3rd party organization
- **PCF validation and automation outlier detection** to ensure Industry Average results are sensible
- **PCF aggregation** where sufficient contributions exist per declared unit
- Final industry average dataset published to LCA database websites and OCP GitHub, **available for free download**.



Call to Action

➔Join the PCR, LCA at Scale and Open Enablers working group

➔Review the LCA at Scale framework

- Estimated publication date: Q1 2027

➔Contribute to the Open Enablers program

- Add PCFs to a long-lived ‘Industry Average’ and open dataset in 2027-8

OCP working group contact

sustainability-product-category@OCP-All.groups.io

Open Enablers timeline

Phase 1: Proof of Concept - working group contributions only

H2 2026 through 2027

Phase 2: Manufacturer PCF contributions

Targeted for 2028

Thank You!

